SG No. 206 T_{h}^{7} $Ia\bar{3}$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$
* Wyckoff site: `8a`, site symmetry: `.-3.`

Table 1: Wyckoff bond: `8a@8a`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$[0,\,0,\,0]$	`[1,5,9,-13,-17,-21]`
2	$[-X,\,-X,\,X]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X,\,X,\,-X]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\,-X,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$	`[4,6,11,-16,-18,-23]`

Table 2: Wyckoff bond: `24b@8a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,0]$	`[1,-13]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[2,-14]`
3	$[-X,\,Y,\,-Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3,-15]`
4	$[X,\,-Y,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$	`[4,-16]`
5	$[Z,\,X,\,Y]$	$[0,\,0,\,0]$	`[5,-17]`
6	$[Z,\,-X,\,-Y]$	$\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$	`[6,-18]`
7	$[-Z,\,-X,\,Y]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[7,-19]`
8	$[-Z,\,X,\,-Y]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[8,-20]`
9	$[Y,\,Z,\,X]$	$[0,\,0,\,0]$	`[9,-21]`
10	$[-Y,\,Z,\,-X]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[10,-22]`
11	$[Y,\,-Z,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$	`[11,-23]`
12	$[-Y,\,-Z,\,X]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `8b`, site symmetry: `.-3.`

Table 3: Wyckoff bond: `8a@8b`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,5,9,-13,-17,-21]`
2	$[-X,\,-X,\,X]$	$\left[\frac{1}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X,\,X,\,-X]$	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{1}{4}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\,-X,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[4,6,11,-16,-18,-23]`

Table 4: Wyckoff bond: `24b@8b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,-13]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[2,-14]`
3	$[-X,\,Y,\,-Z]$	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{1}{4}\right]$	`[3,-15]`
4	$[X,\,-Y,\,-Z]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[4,-16]`
5	$[Z,\,X,\,Y]$	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[5,-17]`
6	$[Z,\,-X,\,-Y]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[6,-18]`

continued ...

Table 4

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,-19]`
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[8,-20]`
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[9,-21]`
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,-22]`
11	$[Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[11,-23]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[12,-24]`

* Wyckoff site: `16c`, site symmetry: `.3.`

Table 5: Wyckoff bond: `16a@16c`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9]`
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[2,7,12]`
3	$[-X, X, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[3,8,10]`
4	$[X, -X, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	`[4,6,11]`
5	$[-X, -X, -X]$	$[-x, -x, -x]$	`[13,17,21]`
6	$[X, X, -X]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[14,19,24]`
7	$[X, -X, X]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[15,20,22]`
8	$[-X, X, X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[16,18,23]`

Table 6: Wyckoff bond: `48b@16c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`
10	$[-Y, Z, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, -x, -x]$	`[13]`
14	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[14]`
15	$[X, -Y, Z]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[15]`
16	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[16]`
17	$[-Z, -X, -Y]$	$[-x, -x, -x]$	`[17]`
18	$[-Z, X, Y]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[18]`
19	$[Z, X, -Y]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[19]`
20	$[Z, -X, Y]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[20]`
21	$[-Y, -Z, -X]$	$[-x, -x, -x]$	`[21]`
22	$[Y, -Z, X]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[22]`

continued ...

Table 6

No.	vector	center	mapping
23	$[-Y, Z, X]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x\right]$	`[23]`
24	$[Y, Z, -X]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-x\right]$	`[24]`

* Wyckoff site: `24d`, site symmetry: `2..`

Table 7: Wyckoff bond: `24a@24d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, 0\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[0, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	`[10,-11]`
7	$[0, -X, -Y]$	$\left[-x, 0, \frac{3}{4}\right]$	`[13,-16]`
8	$[0, X, -Y]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[14,-15]`
9	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, 0\right]$	`[17,-20]`
10	$[-Y, 0, X]$	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	`[18,-19]`
11	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -x\right]$	`[21,-24]`
12	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, x+\frac{1}{2}\right]$	`[22,-23]`

Table 8: Wyckoff bond: `24b@24d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[2,3]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, 0\right]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	`[6,7]`
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, x\right]$	`[9,12]`
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	`[10,11]`
7	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[13,16]`
8	$[X, 0, 0]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[14,15]`
9	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, 0\right]$	`[17,20]`
10	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	`[18,19]`
11	$[0, 0, -X]$	$\left[0, \frac{3}{4}, -x\right]$	`[21,24]`
12	$[0, 0, X]$	$\left[0, \frac{1}{4}, x+\frac{1}{2}\right]$	`[22,23]`

Table 9: Wyckoff bond: `48c@24d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[4]`

continued ...

Table 9

No.	vector	center	mapping
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4},\ x,\ 0\right]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{4},\ x,\ 0\right]$	[8]
9	$[Y,\ Z,\ X]$	$\left[0,\ \frac{1}{4},\ x\right]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[0,\ \frac{1}{4},\ x\right]$	[12]
13	$[-X,\ -Y,\ -Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[13]
14	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[14]
15	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[15]
16	$[-X,\ Y,\ Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[16]
17	$[-Z,\ -X,\ -Y]$	$\left[\frac{3}{4},\ -x,\ 0\right]$	[17]
18	$[-Z,\ X,\ Y]$	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	[18]
19	$[Z,\ X,\ -Y]$	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	[19]
20	$[Z,\ -X,\ Y]$	$\left[\frac{3}{4},\ -x,\ 0\right]$	[20]
21	$[-Y,\ -Z,\ -X]$	$\left[0,\ \frac{3}{4},\ -x\right]$	[21]
22	$[Y,\ -Z,\ X]$	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[22]
23	$[-Y,\ Z,\ X]$	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[23]
24	$[Y,\ Z,\ -X]$	$\left[0,\ \frac{3}{4},\ -x\right]$	[24]

* Wyckoff site: 48e, site symmetry: 1

Table 10: Wyckoff bond: 48a@48e, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[4]
5	$[Z,\ X,\ Y]$	$[z,\ x,\ y]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[8]
9	$[Y,\ Z,\ X]$	$[y,\ z,\ x]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[12]
13	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	[13]
14	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[14]
15	$[X,\ -Y,\ Z]$	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[15]
16	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[16]
17	$[-Z,\ -X,\ -Y]$	$[-z,\ -x,\ -y]$	[17]
18	$[-Z,\ X,\ Y]$	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ y\right]$	[18]
19	$[Z,\ X,\ -Y]$	$\left[z+\frac{1}{2},\ x,\ \frac{1}{2}-y\right]$	[19]
20	$[Z,\ -X,\ Y]$	$\left[z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	[20]
21	$[-Y,\ -Z,\ -X]$	$[-y,\ -z,\ -x]$	[21]

continued ...

Table 10

No.	vector	center	mapping
22	$[Y,\, -Z,\, X]$	$\left[y,\, \frac{1}{2} - z,\, x + \frac{1}{2}\right]$	[22]
23	$[-Y,\, Z,\, X]$	$\left[\frac{1}{2} - y,\, z + \frac{1}{2},\, x\right]$	[23]
24	$[Y,\, Z,\, -X]$	$\left[y + \frac{1}{2},\, z,\, \frac{1}{2} - x\right]$	[24]